

Karthika Sivakumar

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LinkedIn — GitHub — Google Scholar

EDUCATION

Amrita Vishwa Vidyapeetham

M.Tech in Computer Science and Engineering

Coimbatore, India

Expected May 2028

Dr. N.G.P. Institute of Technology

B.Tech in Computer Science and Business Systems

Coimbatore, India

May 2026

- CGPA: 8.78/10.00 — First Class with Distinction.
- Specialization: Honours in Blockchain (Smart Contracts, Solidity, Information Security).

PROJECTS

REOPS AI – Intelligent ML Research Platform

FastAPI, React, Groq LLaMA 3.3, ChromaDB, Scikit-learn

GitHub — Live Deploy

- Engineered a full-stack AI platform supporting 23 ML models with ensemble learning, cross-validation, and automated hyperparameter tuning across 20,000+ OpenML datasets.
- Designed multi-agent LLM orchestration system using Groq LLaMA 3.3 70B and RAG pipeline (ChromaDB) to enable semantic document search and AI-driven experiment analysis.
- Developed an automated ReportLab engine to generate publication-quality research PDFs featuring statistical analysis, feature importance, and model comparisons.
- Implemented CI/CD pipelines on GitHub, deploying a scalable architecture across Render and Vercel.

CRMM-R: On-Device Multimodal Health Reasoning

PyTorch, Edge AI, Model Compression

GitHub — Portfolio

- Engineered a resource-efficient multimodal framework for Medical VQA, optimizing cross-modal semantic interaction via a lightweight cross-attention fusion mechanism.
- Achieved a 7.61% absolute accuracy improvement over baseline concatenation models while maintaining a compact 1.37M parameter footprint.
- Implemented dynamic post-training quantization, reducing model size by over 80% (5.23 MB to 0.90 MB) for deployment on edge-enabled devices.
- Established a holistic evaluation framework assessing reliability, including Expected Calibration Error (ECE of 0.0437) and noise robustness.

Probing Failure Modes of Machine Unlearning in LMs

PyTorch, Hugging Face, DistilGPT-2, Synthetic Benchmarking

GitHub — Portfolio

- Developed a controlled synthetic benchmark to systematically evaluate machine unlearning, addressing limitations in detecting residual memorization within LLMs.
- Proposed a two-stage experimental framework utilizing multi-probe evaluation (direct, paraphrased, indirect, and fill-in-the-blank) to ensure robust verification of knowledge removal.
- Demonstrated a 78% forgetting rate (0.78 to 0.0 target recovery) by implementing an explicit pre-unlearning memorization phase, significantly improving evaluation sensitivity.
- Proved that reliable machine unlearning verification depends on diverse prompting strategies, effectively exposing failure modes where models exhibit apparent forgetting under narrow evaluation.

EXPERIENCE

Pinesphere Solutions

Software Engineering Intern

Coimbatore, India

July 2024 – Aug 2024

- Reduced API response latency by 15% under high-throughput traffic simulations by re-architecting backend business logic pipelines in Django and PostgreSQL.
- Eliminated state synchronization anomalies in distributed front-end systems by optimizing data flow and state management protocols.

RESEARCH & PUBLICATIONS

Green Computing and Energy-Efficient AI Models [2026] – Optimized deep learning inference by 49.2% in energy consumption and reduced carbon emissions by 53.4% using custom pruning and 8-bit quantization algorithms.

ECLNet: Real-Time Intrusion Detection [IEEE] – Achieved 99.2% accuracy in zero-day threat detection by deploying a spatiotemporal fusion CNN-LSTM network in Edge-IoT environments.

Adaptive QoS Optimization for Healthcare IoT [IEEE] – Improved routing accuracy to 94.7% and reduced network congestion by 67% using a multi-agent Deep Q-Network (DQN) pipeline.

Multimodal Fake Video Detection [IEEE] – Developed a cross-modal neural alignment layer to validate authenticity, achieving high classification bounds under heavy lossy compression.

TECHNICAL SKILLS

Languages: Python, SQL.

AI/ML & LLMs: Deep Learning (CNN/LSTM/DQN), LLM Orchestration (Groq, LLaMA), RAG Pipelines, Model Compression (Quantization/Pruning), Computer Vision, NLP.

Frameworks & Databases: Django, FastAPI, React, PostgreSQL, ChromaDB, FAISS, Scikit-learn.

Engineering & DevOps: Distributed Systems, System Architecture Optimization, RESTful API Design, CI/CD (GitHub Actions), Cloud Deployment (Render/Vercel).

ACHIEVEMENTS

GATE Qualified 2026: Scored in Computer Science domain in the Graduate Aptitude Test in Engineering.

Vishwakarma Award: Secured project funding for developing an embedded system for wastewater-based energy generation.

L&T Techgium Finalist: Engineered an end-to-end AI computer vision pipeline for automated content moderation.